VIDEO GAME CREATION

Lesson 7 - Core Mechanics I: On the Move

Semester 1, Week 7 - Core Mechanics unit begins

- **Objectives:** Implement continuous, frame-rate-independent movement; distinguish an object being removed for one reason (a hit) from being removed for another (escaping off-screen); explain why generic "on destroyed" lifecycle hooks can silently misfire once more than one destruction pathway exists; correctly relocate state-changing logic to the specific event that represents it.
- **IGDA Tie-In:** Programming - game loops and continuous simulation, event-driven programming, debugging and code correctness (introductory)
- **Featured Engine (lecture demo):** Godot Engine, contrasted briefly with GameMaker for the bug discussion
- **Materials:** Demo laptop with all 8 engines verified (see Teacher's Checklist); Student Handouts, one per engine; students' own Lesson 6 projects (already saved) - no new assets needed this week

Agenda

0:00-0:05	Welcome back, Lesson 6 recap, milestone check-in
0:05-0:20	New Unit: Core Mechanics - what's ahead through Lesson 11
0:20-0:40	Movement That Doesn't Stop: Continuous Motion (live demo in Godot)
0:40-1:00	The Hidden Bug: When 'Destroyed' Doesn't Mean 'Defeated'
1:00-1:05	Introduce lab exercise, hand out engine-specific handouts
1:05-1:55	LAB: Advance enemies toward the player + escape cleanup
1:55-2:00	Exit check / quick share-out

Levels for this lesson's exercise

- Level 1 (Minimal): every enemy moves slowly downward; an enemy that reaches the bottom unshot is removed without counting as a kill.
- Level 2 (Complete): a moving enemy can still be shot correctly, and an escaped enemy never triggers "You Win!" by accident.
- Level 3 (Refined): give enemies varied speeds, and/or tighten the separation between movement/escape code and hit-detection/win-counting code.

See: Lesson 7 Teacher's Script, Lesson 7 Prep Checklist, and the per-engine Student Handouts. GameMaker and 001 Game Creator instructors: read the Prep Checklist's bugfix note before class.